HW3

Group 12 - Shir, Omer & Ido

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Group:

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Paths and Data

```r
if (!require('data.table'))  { install.packages('data.table');  library('data.table') }
```

```
## Loading required package: data.table
```

```r
if (!require('dplyr'))       { install.packages('dplyr');       library('dplyr') }
```

```
## Loading required package: dplyr

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:data.table':
##
##     between, first, last

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
if (!require('kableExtra'))  { install.packages('kableExtra');  library('kableExtra') }
```

```
## Loading required package: kableExtra

##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```r
reads_file  <- "../lab2/data/TCGA-13-0723-10B_lib2_all_chr1.forward"
beg_region  <- 50000000
end_region  <- 60000000
```

Reading the Data:

```r
chr1_reads <- fread(reads_file, col.names = c("Chr", "Loc", "Length"))
```

Introduction

In this lab we model base-level read coverage using the Poisson distribution and assess how well it fits the data.

Part 1

Question 1

Computing Lambda — mean coverage per base:

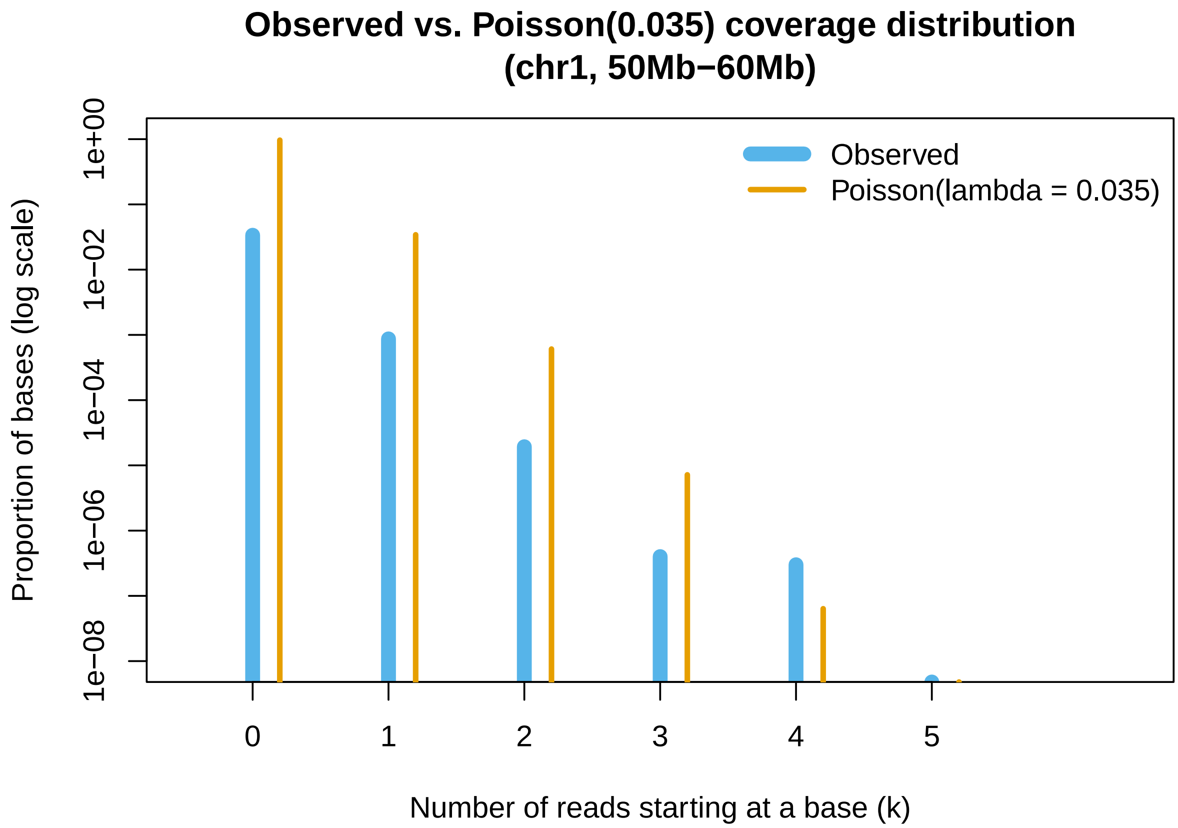
```r
source("../utils.R")

N      <- end_region - beg_region + 1
cov    <- count_reads_table(chr1_reads, beg_region, end_region)
lambda <- mean(cov)
cat(sprintf("lambda = %.4f\n", lambda))
```

```
## lambda = 0.0355
```

Expected Poisson distribution and comparison:

```r
# observed proportions
obs_counts <- tabulate(cov, nbins = max(cov) + 1)   # index k+1 = count of k
obs_prop   <- obs_counts / N

# expected Poisson probabilities
```

```r
k_vals   <- 0:max(cov)
exp_prop <- dpois(k_vals, lambda)

# --- Graph ---
x_max    <- 6
k_show   <- k_vals[k_vals <= x_max]
obs_show <- pmax(obs_prop[k_show + 1], 1e-10)   # avoid log(0)
exp_show <- pmax(exp_prop[k_show + 1], 1e-10)

y_min <- 10^floor(log10(min(obs_show[obs_show > 1e-9], exp_show[exp_show > 1e-9])))

plot(k_show, obs_show,
     type = "h", lwd = 8, col = "#56B4E9", log = "y",
     xlab = "Number of reads starting at a base (k)",
     ylab = "Proportion of bases (log scale)",
     main = sprintf("Observed vs. Poisson(%s) coverage distribution\n(chr1,
      ↪ %dMb\u2013%dMb)",
                    signif(lambda, 2), beg_region/1e6, end_region/1e6),
     xlim = c(-0.5, x_max + 0.5),
     ylim = c(y_min, 1),
     xaxt = "n")
axis(1, at = k_show)
lines(k_show + 0.2, exp_show, type = "h", lwd = 3, col = "#E69F00")
legend("topright",
       legend = c("Observed", sprintf("Poisson(lambda = %s)", signif(lambda, 2))),
       col    = c("#56B4E9", "#E69F00"),
       lwd    = c(8, 3),
       bty    = "n")
```



The graph shows the expected poisson reads histogram compared to the observed histogram. Before we used log-scale it was hard to distinguish exactly since the distribution with lambda =0.036 is skewed. We can see that Poisson distribution does not really match the actual data, which is less left-skewed and more right-skewed.

```
obs_vals <- c(sapply(0:5, function(k) sum(cov == k) / N), sum(cov > 5) / N)
exp_vals <- c(dpois(0:5, lambda), ppois(5, lambda, lower.tail = FALSE))

# fixed decimal places: enough to show variation (all values are small proportions)
fmt <- function(x) formatC(x, format = "f", digits = 7)

data.frame(k = c(as.character(0:5), ">5"),
           Observed = fmt(obs_vals),
           Expected = fmt(exp_vals)) %>%
  knitr::kable(align = "c",
               caption = sprintf("Observed vs. Expected Poisson proportions (lambda =
                ↪ %s)",
                                 signif(lambda, 2))) %>%
  kable_styling(bootstrap_options = c("bordered", "striped"), full_width = FALSE) %>%
  column_spec(1:3, border_left = TRUE, border_right = TRUE)
```

Table 1: Observed vs. Expected Poisson proportions (lambda = 0.035)

k	Observed	Expected
0	0.9654511	0.9651629
1	0.0336604	0.0342231
2	0.0008685	0.0006067
3	0.0000193	0.0000072
4	0.0000004	0.0000001
5	0.0000003	0.0000000
>5	0.0000000	0.0000000

It's harder for us to notice the difference in the table compared to the graph, but from the graph it's clear that the Poisson distribution is not a very good fit.

Question 2

```
BIN_SIZE <- 5000

bin_counts <- sapply(seq(1, N - BIN_SIZE + 1, by = BIN_SIZE), function(i) {
  sum(cov[i:(i + BIN_SIZE - 1)])
}) # for each bin, we apply for the sum function over cov vector.

lambda_bin <- lambda * BIN_SIZE  # expected mean under Poisson
```

a. Robust estimators vs. Poisson prediction:

```
med    <- median(bin_counts)
iqr    <- IQR(bin_counts)
# For Poisson, IQR   1.35 * sqrt(lambda) (normal approximation)
exp_iqr <- qpois(0.75, lambda_bin) - qpois(0.25, lambda_bin)

cat(sprintf("Median:       %.2f  (Poisson mean: %.2f)\n", med, lambda_bin))
```

```
## Median:       179.00  (Poisson mean: 177.29)
```

```
cat(sprintf("IQR:          %.2f  (Poisson expected IQR   %.2f)\n", iqr, exp_iqr))
```

```
## IQR:          73.00  (Poisson expected IQR   18.00)
```

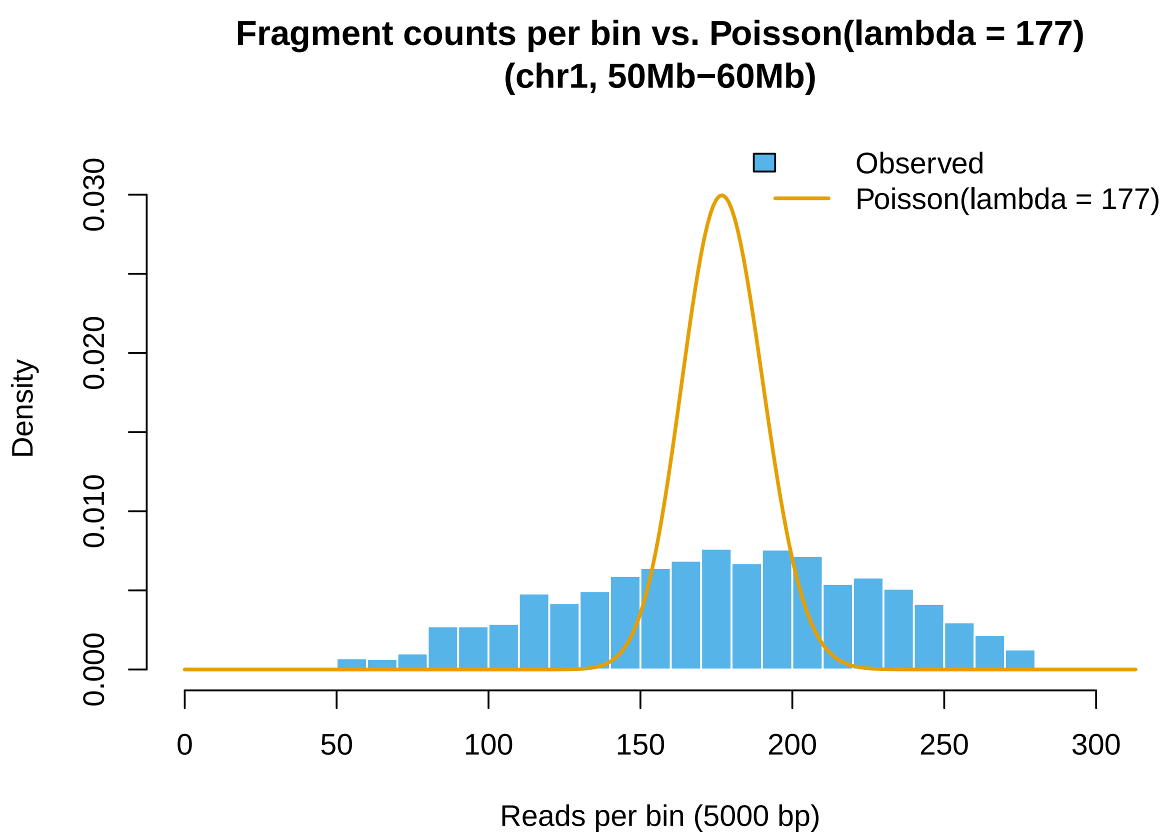
As we've seen in class, we use the median as a robust estimate of the center instead of the average. For spread, we use IQR as a robust estimator.

We can see that the center is pretty close to the observed distribution with a small bias. The spread, however, is almost 5 times bigger than we expected under the Poisson model.

b. Histogram vs. Poisson:

```
x_max_b   <- quantile(bin_counts, 0.99)
bins_trim <- bin_counts[bin_counts <= x_max_b]

h <- hist(bins_trim, breaks = 30, plot = FALSE)
```

```r
# smooth Poisson curve: dpois at every integer in range
k_range   <- 0:max(bin_counts)
pois_dens <- dpois(k_range, lambda_bin)

plot(h, freq = FALSE,
     col = "#56B4E9", border = "white",
     xlim = c(min(h$breaks), max(bin_counts)),
     ylim = c(0, max(h$density, pois_dens) * 1.1),
     xlab = sprintf("Reads per bin (%d bp)", BIN_SIZE),
     ylab = "Density",
     main = sprintf("Fragment counts per bin vs. Poisson(lambda = %s)\n(chr1,
     ↪  %dMb\u2013%dMb)",
                    signif(lambda_bin, 3), beg_region/1e6, end_region/1e6))
lines(k_range, pois_dens, col = "#E69F00", lwd = 2)

legend("topright",
       legend = c("Observed", sprintf("Poisson(lambda = %s)", signif(lambda_bin, 3))),
       fill   = c("#56B4E9", NA), border = c("black", NA),
       lty    = c(NA, 1), col = c(NA, "#E69F00"), lwd = c(NA, 2),
       bty    = "n")
```



In the lambda value we got, 177, the expected distribution looks almost identical to a normal distribution. As we mentioned in a, the center is aligned with the expectation, but the observed data spread is much higher, indicating once again that the Poisson model is not a good fit.

c. Variance-to-Mean Ratio (VMR):

```
vmr <- var(bin_counts) / mean(bin_counts)
cat(sprintf("Mean:     %.4f\n", mean(bin_counts)))
```

```
## Mean:     177.2920
```

```
cat(sprintf("Variance: %.4f\n", var(bin_counts)))
```

```
## Variance: 2605.7396
```

```
cat(sprintf("VMR:      %.4f  (Poisson expectation: 1)\n", vmr))
```

```
## VMR:      14.6974  (Poisson expectation: 1)
```

For a Poisson distribution, VMR = 1 by definition (variance equals mean). The VMR we actually got, 14.6974, again indicates this model is not a good choice for our specific case. Perhaps one of our assumptions - uniformity or independence is incorrect.

Question 3

```
load("../lab1/data/chr1_str.rda")

gc_beg <- 50000000
gc_end <- 100000000

# coverage for the full 50Mb region
cov_50mb <- count_reads_table(chr1_reads, gc_beg, gc_end)

# bin coverage
bin_counts_50mb <- sapply(seq(1, length(cov_50mb) - BIN_SIZE + 1, by = BIN_SIZE),
 ↪  function(i) {
  sum(cov_50mb[i:(i + BIN_SIZE - 1)])
})

# GC for the same full 50Mb region
seq_region <- substr(chr1, gc_beg, gc_end)
split_seq  <- strsplit(seq_region, "")[[1]]

gc_binned_50mb <- sapply(seq(1, length(split_seq) - BIN_SIZE + 1, by = BIN_SIZE),
 ↪  function(i) {
  sum(split_seq[i:(i + BIN_SIZE - 1)] %in% c("C", "G", "c", "g"))
})

# align lengths just in case
min_len <- min(length(bin_counts_50mb), length(gc_binned_50mb))

cov_sub <- bin_counts_50mb[1:min_len]
gc_sub  <- gc_binned_50mb[1:min_len]
```

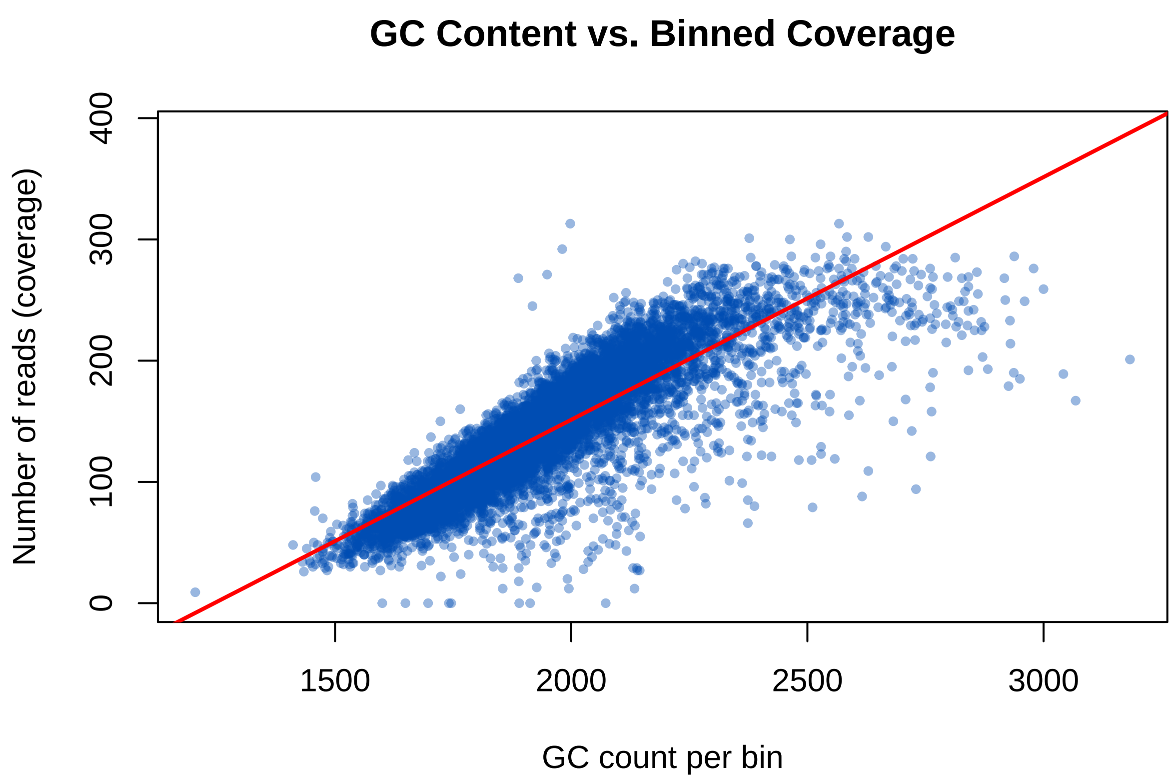
a. Calculate correlation

```
cor_val <- cor(gc_sub, cov_sub)
cat(sprintf("Correlation between GC content and coverage: %.4f\n", cor_val))
```

```
## Correlation between GC content and coverage: 0.8351
```

The calculated correlation coefficient is 0.8351, showing a strong positive linear relationship between the GC content and the observed read coverage per bin. This direction is consistent with the findings in Dohm paper, who reported that read coverage was biased toward GC-rich regions, with increased read density in intervals of elevated GC content. Our result supports the idea that coverage is not uniform across the genome, but is affected by local sequence composition.

```
y_cap <- 390
plot(gc_sub, cov_sub,
     xlab = "GC count per bin",
     ylab = "Number of reads (coverage)",
     main = "GC Content vs. Binned Coverage",
     pch = 20, col = rgb(0, 0.3, 0.7, 0.4), cex = 0.8, ylim= c(0,y_cap))
abline(lm(cov_sub ~ gc_sub), col = "red", lwd = 2)
```



From the plot we can observe that there is a strong positive linear relation between the GC content and the read coverage. This is similar to the trend reported by Dohm et al., where read coverage increased in regions with elevated GC content. In our plot, the relationship appears approximately linear over the observed GC range, while Dohm's figures focus more generally on demonstrating GC-related coverage bias. In both cases, the main conclusion is that sequencing coverage is not uniform and is influenced by GC content.

c. Relationship to Part 1 The strong positive correlation between GC content and coverage explains the large spread and high variance-mean ratio observed in Part 1. The assumption of a uniform Poisson distribution doesn't hold since read coverage is not random or even across the chromosome. Instead, it is driven, to some extent, by local GC content.

Short summary

In the first part we tried to asses the Poisson model fit for our data, and saw it has significant inconsistencies with the observed distribution.

In the second part, we used genomic binning to looked at the correlation between GC content and read coverage.

External sources statement

We also used AI (Claude) to:

1. move the function from lab2 to a utils file
2. edit the graph legend in q1.
3. use `knitr::kable` for table creation in q1.
4. correctly use `qpois` to calculate the expected IQR